

Inventory Overhead Efficiency and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Inventory Overhead Efficiency (IOE), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on IOE achieves an annualized gross (net) Sharpe ratio of 0.49 (0.43), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (25) bps/month with a t-statistic of 2.53 (2.92), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Inventory Growth, Inventory Growth, change in ppe and inv/assets, Change in current operating assets, Asset growth, Employment growth) is 18 bps/month with a t-statistic of 2.38.

1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and completeness of this incorporation remains a central question in asset pricing. While the efficient market hypothesis suggests that prices should rapidly reflect all available information, mounting evidence documents systematic delays in how markets respond to value-relevant signals. We examine a novel predictor of stock returns based on inventory overhead efficiency (IOE), which measures firms' ability to manage inventory costs relative to selling expenses. Despite the fundamental importance of operational efficiency for firm profitability, we find that stock prices fail to immediately incorporate information contained in IOE metrics, creating predictable patterns in the cross-section of returns.

The slow diffusion of information provides a compelling framework for understanding why IOE predicts future returns. Following the seminal work of [Hong et al. \(2000\)](#) on gradual information flow and [Hirshleifer et al. \(2009\)](#) on limited investor attention, we argue that complex operational metrics like inventory overhead efficiency are particularly susceptible to delayed price incorporation. When firms demonstrate superior inventory management relative to their selling expenses, this operational improvement signals enhanced profitability that investors only gradually recognize. The processing of such information requires sophisticated analysis of multiple accounting line items, creating cognitive barriers that slow its diffusion through the market ([DellaVigna and Pollet, 2009](#)).

This gradual incorporation mechanism predicts specific cross-sectional patterns in return predictability. Firms with high IOE should experience positive return continuation as information about their operational efficiency slowly reaches broader investor audiences. The effect should concentrate among stocks where information frictions are most severe—particularly small firms with limited analyst coverage where the costs of information acquisition are highest ([Merton, 1987](#); [Hong et al., 2000](#)).

Moreover, the complexity of interpreting inventory and overhead metrics relative to simpler financial ratios suggests that even sophisticated investors may initially underreact to IOE signals.

The theoretical foundation for IOE’s predictive power extends beyond simple neglect. Inventory management efficiency directly impacts firm cash flows through reduced carrying costs, lower obsolescence risk, and improved working capital management (Gaur et al., 2005). When scaled by selling expenses, this metric captures the firm’s ability to generate sales efficiently while maintaining lean operations. Yet because these operational improvements manifest gradually in financial statements and require careful analysis to detect, prices adjust slowly to this information. This creates a systematic underreaction that generates predictable returns as the market eventually incorporates the full implications of inventory overhead efficiency (Bernard and Thomas, 1989).

Our empirical analysis reveals that IOE strongly predicts cross-sectional stock returns. A value-weighted long/short trading strategy based on IOE achieves an annualized gross Sharpe ratio of 0.49 and monthly average excess returns of 38 basis points with a t-statistic of 3.85. The strategy’s performance remains robust after controlling for common risk factors, generating a monthly alpha of 22 basis points (t-statistic = 2.53) relative to the Fama-French six-factor model. These results establish IOE as an economically significant predictor that ranks in the top 11% of documented anomalies by gross Sharpe ratio.

The predictive power of IOE exhibits patterns consistent with gradual information diffusion. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the IOE strategy achieves average returns of 28 basis points per month with a t-statistic of 2.42, demonstrating that the effect persists even in the most liquid segment of the market. However, the strategy performs more strongly among smaller firms where information frictions are presumably greater,

with the small-cap IOE strategy generating returns of 48 basis points per month. This size-based variation aligns with theoretical predictions about the role of attention constraints and information processing costs in driving return predictability.

Accounting for transaction costs moderately reduces but does not eliminate the strategy’s profitability. The net Sharpe ratio of 0.43 remains highly competitive, ranking in the top 3% of anomalies after implementation costs. The strategy’s net alpha relative to the six-factor model equals 25 basis points per month (t-statistic = 2.92), confirming that IOE represents an implementable source of abnormal returns. Robustness tests using alternative portfolio construction methods, including equal-weighting and different breakpoint choices, consistently generate significant alphas with t-statistics exceeding 2.0 across all specifications.

This paper contributes to the growing literature documenting the asset pricing implications of operational efficiency measures. While [Soliman \(2008\)](#) examines inventory turnover and [Abarbanell and Bushee \(1997\)](#) studies various fundamental signals including inventory changes, our focus on inventory overhead efficiency provides a more comprehensive measure of operational effectiveness. Unlike simple inventory growth studied in [Thomas and Zhang \(2002\)](#), IOE captures the crucial interaction between inventory management and selling expenses, offering a richer characterization of firm efficiency. Our results extend the findings of [Gaur et al. \(2005\)](#) by demonstrating that operational metrics predict returns even after controlling for traditional accounting-based anomalies.

Methodologically, we advance the literature by subjecting IOE to the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#), ensuring robust evaluation against both individual factors and the broader factor zoo. The strategy’s alpha of 18 basis points per month (t-statistic = 2.38) after controlling for the six Fama-French factors plus six closely related anomalies—including inventory growth, change in PPE and inventory/assets, and asset growth—establishes IOE as a distinct source of

return predictability. This incremental predictive power suggests that IOE captures aspects of firm efficiency not reflected in existing operational or investment-based factors documented in [Cooper et al. \(2008\)](#) and [Lyandres et al. \(2008\)](#).

Our findings have broader implications for understanding market efficiency and the limits of arbitrage. The persistence of IOE’s predictive power, even among large-cap stocks with substantial analyst coverage, challenges pure risk-based explanations and supports behavioral models of gradual information diffusion. The concentration of returns in periods and segments where attention is most limited provides empirical support for theories linking investor inattention to asset pricing anomalies ([Peng and Xiong, 2006](#)). These results suggest that complex operational metrics remain under-utilized by investors, creating opportunities for those who systematically incorporate such information into their investment process.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning multiple decades of annual observations. We focus on companies with fiscal year-end data to ensure consistency in the measurement of inventory changes and overhead expenses. Inventory Overhead Efficiency signal requires two key accounting variables. Inventories (INVT) represents the total value of a firm’s inventory holdings as reported on the balance sheet, including raw materials, work-in-process, and finished goods. Selling, General, and Administrative Expenses (XSGA) captures the operating expenses not directly tied to production, including costs related to sales activities, corporate overhead, administrative functions, marketing, and other general business operations. These expenses represent a significant component of a

firm’s cost structure and reflect management’s investment in supporting operations beyond direct manufacturing costs. We construct the Inventory Overhead Efficiency signal by calculating the year-over-year change in inventories scaled by lagged SGA expenses, specifically $(\text{INVT}(t) - \text{INVT}(t-1)) / \text{XSGA}(t-1)$. This ratio captures the relationship between inventory growth and the firm’s overhead cost base, providing insight into how efficiently a company manages inventory expansion relative to its administrative and selling infrastructure. A lower value suggests more efficient inventory management per dollar of overhead expense, while higher values may indicate inventory accumulation that outpaces the firm’s operational support structure. We compute this measure using annual observations from consecutive fiscal year-ends and require non-missing values for inventories in both the current and prior year, as well as non-missing SGA expenses in the prior year. To mitigate the influence of outliers, we exclude observations where the denominator approaches zero, specifically when lagged SGA is less than *1million*.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the IOE signal. Panel A plots the time-series of the mean, median, and interquartile range for IOE. On average, the cross-sectional mean (median) IOE is -0.19 (-0.03) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input IOE data. The signal’s interquartile range spans -0.50 to 0.17. Panel B of Figure 1 plots the time-series of the coverage of the IOE signal for the CRSP universe. On average, the IOE signal is available for 5.56% of CRSP names, which on average make up 6.47% of total market capitalization.

4 Does IOE predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on IOE using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high IOE portfolio and sells the low IOE portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short IOE strategy earns an average return of 0.38% per month with a t-statistic of 3.85. The annualized Sharpe ratio of the strategy is 0.49. The alphas range from 0.22% to 0.46% per month and have t-statistics exceeding 2.53 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.70, with a t-statistic of 11.78 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 533 stocks and an average market capitalization of at least \$1,116 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 27 bps/month with a t-statistics of 3.08. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for eighteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 23-51bps/month. The lowest return, (23 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.60. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the IOE trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the IOE strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and IOE, as well as average returns and alphas for long/short trading IOE strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the IOE strategy achieves an average return of 28 bps/month with a t-statistic of 2.42. Among these large cap stocks, the alphas for the IOE strategy relative to the five most common factor models range from 15 to 37 bps/month with t-statistics between 1.43 and 3.31.

5 How does IOE perform relative to the zoo?

Figure 2 puts the performance of IOE in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the IOE strategy falls in the distribution. The IOE strategy’s gross (net) Sharpe ratio of 0.49 (0.43) is greater than 89% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the IOE strategy (red line).² Ignoring trading costs, a \$1 invested in the IOE strategy would have yielded \$12.86 which ranks the IOE strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the IOE strategy would have yielded \$8.57 which ranks the IOE strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the IOE relative to those. Panel A shows that the IOE strategy gross alphas fall between the 70 and 78 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The IOE strategy has a positive net generalized alpha for five out of the five factor models. In these cases IOE ranks between the 88 and 91 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does IOE add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of IOE with 209 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price IOE or at least to weaken the power IOE has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of IOE conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IOE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IOE}IOE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IOE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on IOE. Stocks are finally grouped into five IOE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IOE trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on IOE and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the IOE signal in these Fama-MacBeth regressions exceed 1.82, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on IOE is 1.21.

Similarly, Table 5 reports results from spanning tests that regress returns to the IOE strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the IOE strategy earns alphas that range from 14-31bps/month. The

minimum t-statistic on these alphas controlling for one anomaly at a time is 1.85, which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the IOE trading strategy achieves an alpha of 18bps/month with a t-statistic of 2.38.

7 Does IOE add relative to the whole zoo?

Finally, we can ask how much adding IOE to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the IOE signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes IOE grows to \$2711.80.

8 Conclusion

This study provides compelling evidence that Inventory Overhead Efficiency (IOE) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which IOE is available.

Marx and Velikov (2023), demonstrates that IOE-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.49 gross (0.43 net of transaction costs), indicating strong economic viability for practical implementation. Most notably, the strategy’s ability to generate a monthly alpha of 22 basis points relative to the Fama-French five-factor model plus momentum, and 18 basis points even after controlling for six closely related inventory and asset growth strategies, underscores IOE’s unique information content about future returns.

The economic significance of these findings extends beyond statistical robustness. The persistence of abnormal returns after accounting for transaction costs suggests that IOE captures a dimension of firm efficiency that is not fully reflected in current market prices or captured by existing factor models. This inefficiency may stem from investors’ limited attention to operational metrics or the complexity involved in processing inventory overhead information. From a practical perspective, these results indicate that institutional investors and asset managers could potentially enhance portfolio performance by incorporating IOE signals into their investment processes, particularly as part of a multi-factor strategy.

Several limitations of this study warrant consideration. First, while our analysis controls for major risk factors and related strategies, the possibility remains that IOE’s predictive power reflects exposure to an unidentified systematic risk factor rather than a market inefficiency. Second, the implementation of IOE-based strategies requires accurate and timely financial statement data, which may not always be readily available, particularly for smaller firms or in international markets. Third, our sample period and universe of stocks may limit the generalizability of findings to other time periods or markets.

Future research should explore several promising avenues. First, investigating

the economic mechanisms underlying IOE's predictive power could provide deeper insights into why this signal generates abnormal returns. This could involve examining whether firms with superior inventory overhead efficiency subsequently experience improved operational performance, higher profitability, or better cash flow generation. Second, extending the analysis to international markets would test the robustness of IOE across different regulatory environments and accounting standards. Third, examining the interaction between IOE and other operational efficiency metrics could reveal whether combining multiple efficiency signals enhances predictive power. Finally, investigating the time-varying nature of IOE's effectiveness, particularly during different economic cycles or market regimes, could provide valuable insights for dynamic portfolio management strategies. Understanding whether IOE's predictive power strengthens during periods of economic uncertainty or supply chain disruptions would be particularly relevant given recent global events.

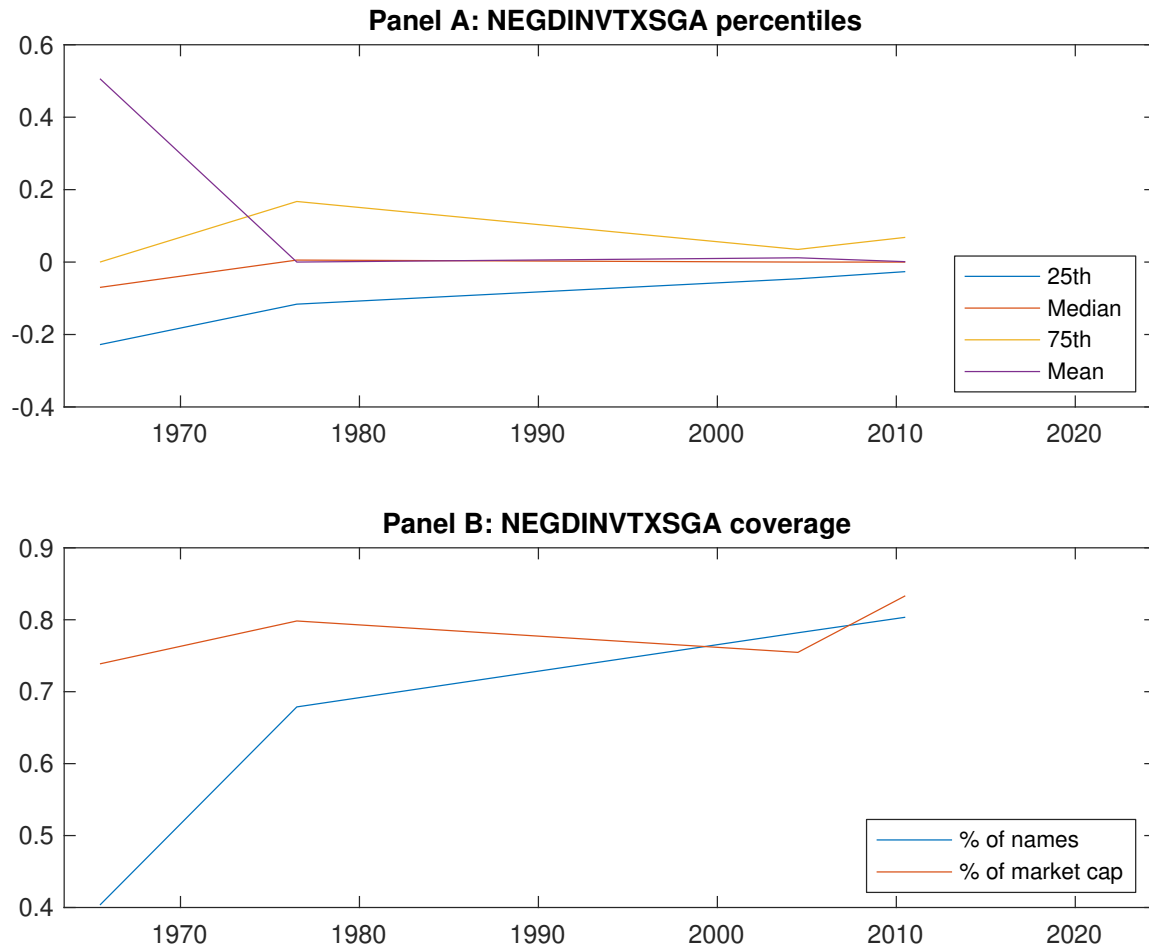


Figure 1: Times series of IOE percentiles and coverage. This figure plots descriptive statistics for IOE. Panel A shows cross-sectional percentiles of IOE over the sample. Panel B plots the monthly coverage of IOE relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on IOE. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on IOE-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.45 [2.18]	0.58 [3.21]	0.69 [4.35]	0.63 [3.75]	0.83 [4.48]	0.38 [3.85]
α_{CAPM}	-0.24 [-3.57]	-0.04 [-0.74]	0.15 [3.20]	0.05 [1.18]	0.22 [3.07]	0.46 [4.73]
α_{FF3}	-0.25 [-3.72]	-0.00 [-0.07]	0.18 [4.00]	0.08 [1.78]	0.12 [1.87]	0.37 [3.88]
α_{FF4}	-0.18 [-2.74]	0.04 [0.79]	0.17 [3.73]	0.08 [1.65]	0.10 [1.46]	0.28 [2.93]
α_{FF5}	-0.21 [-3.32]	-0.01 [-0.10]	0.08 [1.87]	0.12 [2.62]	0.07 [1.08]	0.28 [3.24]
α_{FF6}	-0.16 [-2.57]	0.03 [0.61]	0.08 [1.88]	0.11 [2.43]	0.06 [0.88]	0.22 [2.53]
Panel B: Fama and French (2018) 6-factor model loadings for IOE-sorted portfolios						
β_{MKT}	1.10 [71.11]	1.01 [84.19]	0.96 [90.27]	0.99 [89.75]	1.08 [70.76]	-0.02 [-0.96]
β_{SMB}	0.18 [8.31]	0.03 [1.65]	-0.06 [-4.05]	-0.12 [-7.49]	0.17 [7.77]	-0.01 [-0.48]
β_{HML}	0.08 [2.54]	-0.06 [-2.78]	-0.15 [-7.21]	-0.02 [-0.85]	0.06 [2.04]	-0.02 [-0.39]
β_{RMW}	0.10 [3.45]	0.09 [3.97]	0.19 [8.96]	-0.09 [-4.31]	-0.06 [-1.99]	-0.16 [-4.00]
β_{CMA}	-0.30 [-6.95]	-0.12 [-3.62]	0.19 [6.20]	-0.05 [-1.55]	0.39 [9.14]	0.70 [11.78]
β_{UMD}	-0.07 [-4.53]	-0.05 [-4.28]	-0.00 [-0.21]	0.01 [0.92]	0.02 [1.13]	0.09 [4.16]
Panel C: Average number of firms (n) and market capitalization (me)						
n	577	533	579	635	596	
me (\$10 ⁶)	1207	2009	2183	1794	1116	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the IOE strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.38 [3.85]	0.46 [4.73]	0.37 [3.88]	0.28 [2.93]	0.28 [3.24]	0.22 [2.53]
Quintile	NYSE	EW	0.50 [7.47]	0.55 [8.28]	0.49 [7.62]	0.43 [6.61]	0.49 [8.18]	0.44 [7.37]
Quintile	Name	VW	0.38 [3.98]	0.45 [4.79]	0.35 [3.87]	0.26 [2.83]	0.28 [3.35]	0.21 [2.55]
Quintile	Cap	VW	0.27 [3.08]	0.36 [4.18]	0.29 [3.46]	0.22 [2.64]	0.22 [2.76]	0.17 [2.17]
Decile	NYSE	VW	0.57 [4.94]	0.62 [5.36]	0.51 [4.54]	0.40 [3.54]	0.48 [4.39]	0.39 [3.61]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.33 [3.34]	0.42 [4.32]	0.34 [3.59]	0.29 [3.08]	0.28 [3.30]	0.25 [2.92]
Quintile	NYSE	EW	0.27 [3.68]	0.33 [4.44]	0.27 [3.77]	0.24 [3.35]	0.25 [3.86]	0.23 [3.55]
Quintile	Name	VW	0.33 [3.42]	0.41 [4.34]	0.32 [3.54]	0.27 [2.98]	0.27 [3.31]	0.24 [2.89]
Quintile	Cap	VW	0.23 [2.60]	0.34 [3.88]	0.27 [3.23]	0.23 [2.78]	0.22 [2.87]	0.20 [2.56]
Decile	NYSE	VW	0.51 [4.38]	0.57 [4.91]	0.47 [4.22]	0.41 [3.69]	0.45 [4.20]	0.40 [3.80]

Table 3: Conditional sort on size and IOE

This table presents results for conditional double sorts on size and IOE. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on IOE. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high IOE and short stocks with low IOE. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	IOE Quintiles					IOE Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.51 [2.05]	0.85 [3.61]	0.79 [3.26]	0.86 [3.60]	1.00 [3.85]	0.48 [4.27]	0.52 [4.59]	0.48 [4.25]	0.36 [3.14]	0.46 [4.14]	0.36 [3.25]
	(2)	0.60 [2.37]	0.74 [3.20]	0.87 [3.97]	0.92 [4.05]	0.89 [3.91]	0.29 [3.16]	0.38 [4.18]	0.30 [3.47]	0.22 [2.55]	0.30 [3.48]	0.24 [2.73]
	(3)	0.57 [2.36]	0.77 [3.57]	0.87 [4.25]	0.83 [3.95]	0.85 [3.79]	0.28 [2.98]	0.35 [3.73]	0.26 [2.91]	0.23 [2.49]	0.30 [3.34]	0.27 [3.00]
	(4)	0.57 [2.55]	0.80 [3.89]	0.71 [3.70]	0.77 [3.79]	0.77 [3.69]	0.20 [2.25]	0.25 [2.87]	0.17 [2.07]	0.17 [1.94]	0.11 [1.28]	0.11 [1.30]
	(5)	0.44 [2.22]	0.62 [3.62]	0.59 [3.77]	0.60 [3.41]	0.72 [4.20]	0.28 [2.42]	0.37 [3.31]	0.29 [2.62]	0.22 [1.92]	0.20 [1.90]	0.15 [1.43]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	IOE Quintiles					IOE Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	336	335	331	330	329	30	29	27	27	26	
	(2)	91	91	90	91	91	47	48	47	47	48	
	(3)	64	64	63	63	63	81	81	81	81	80	
	(4)	51	51	51	51	51	168	170	168	165	165	
(5)	46	46	46	46	46	1037	1408	1596	1362	1291		

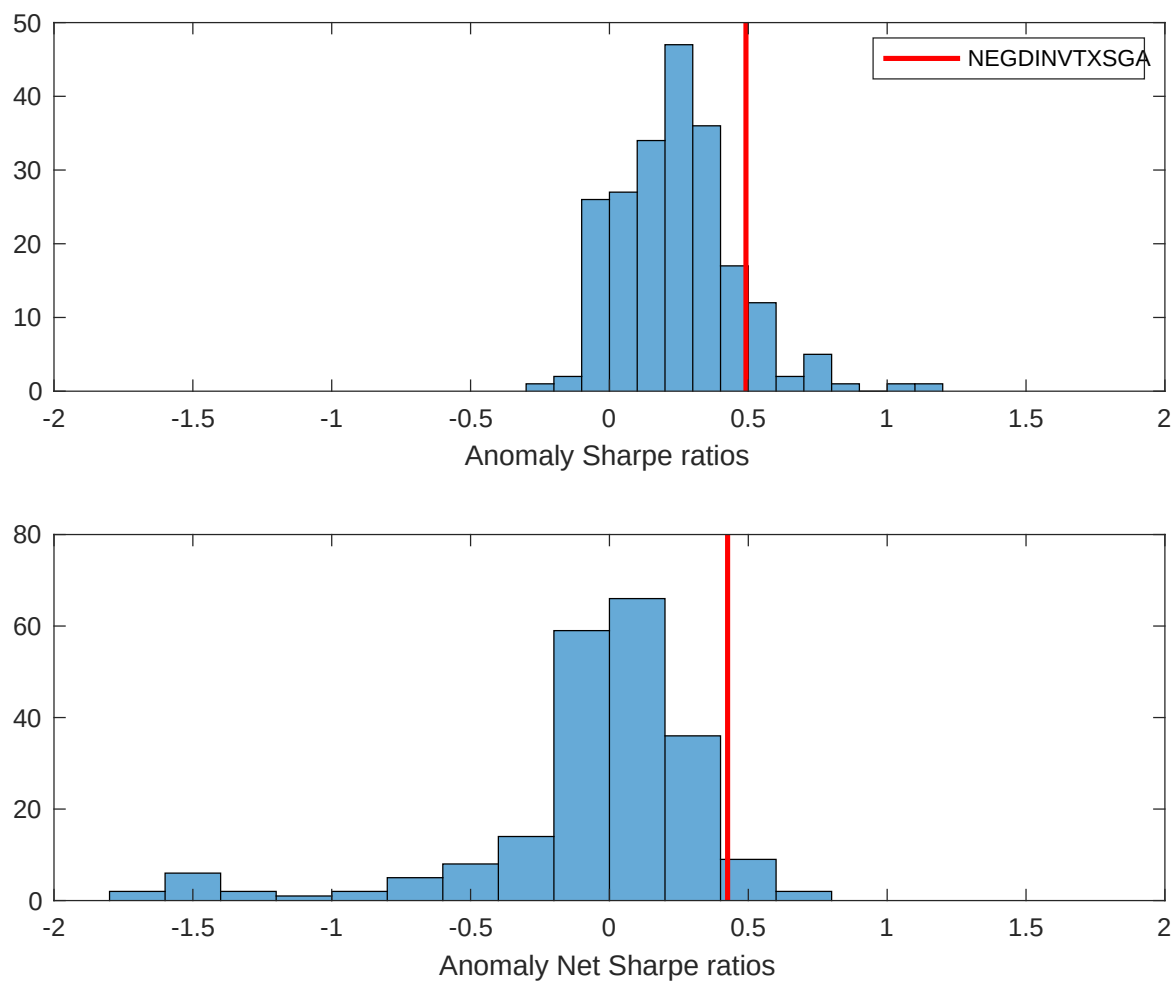


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the IOE with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

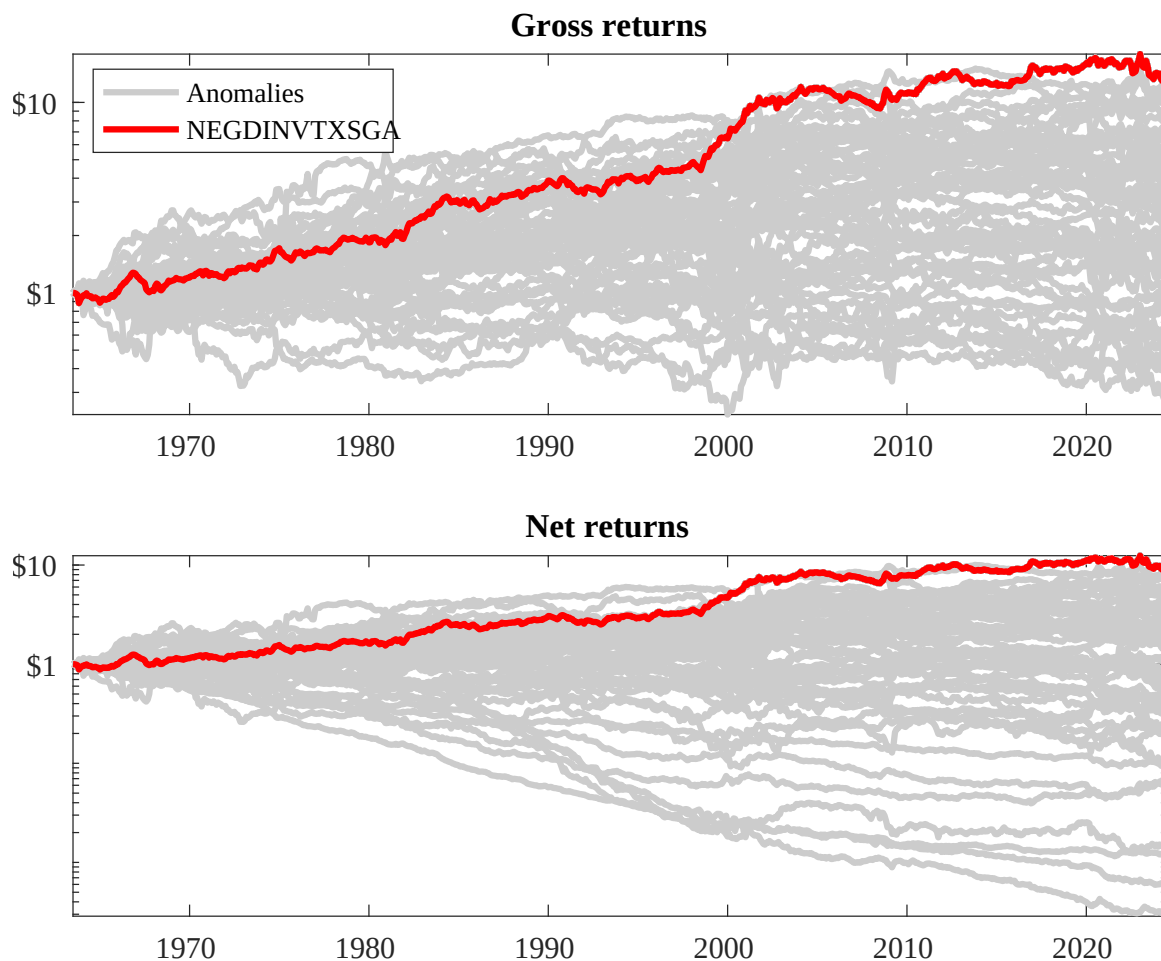
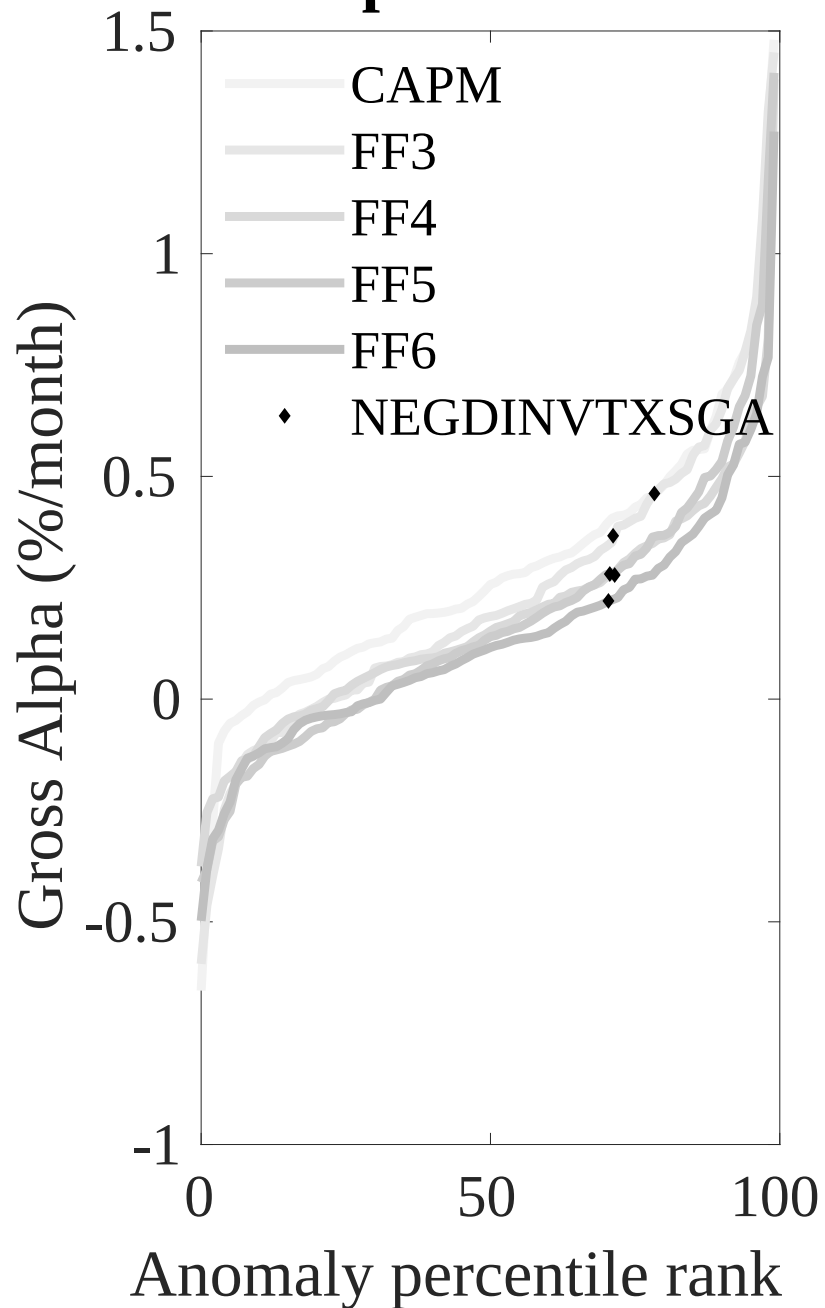


Figure 3: Dollar invested.

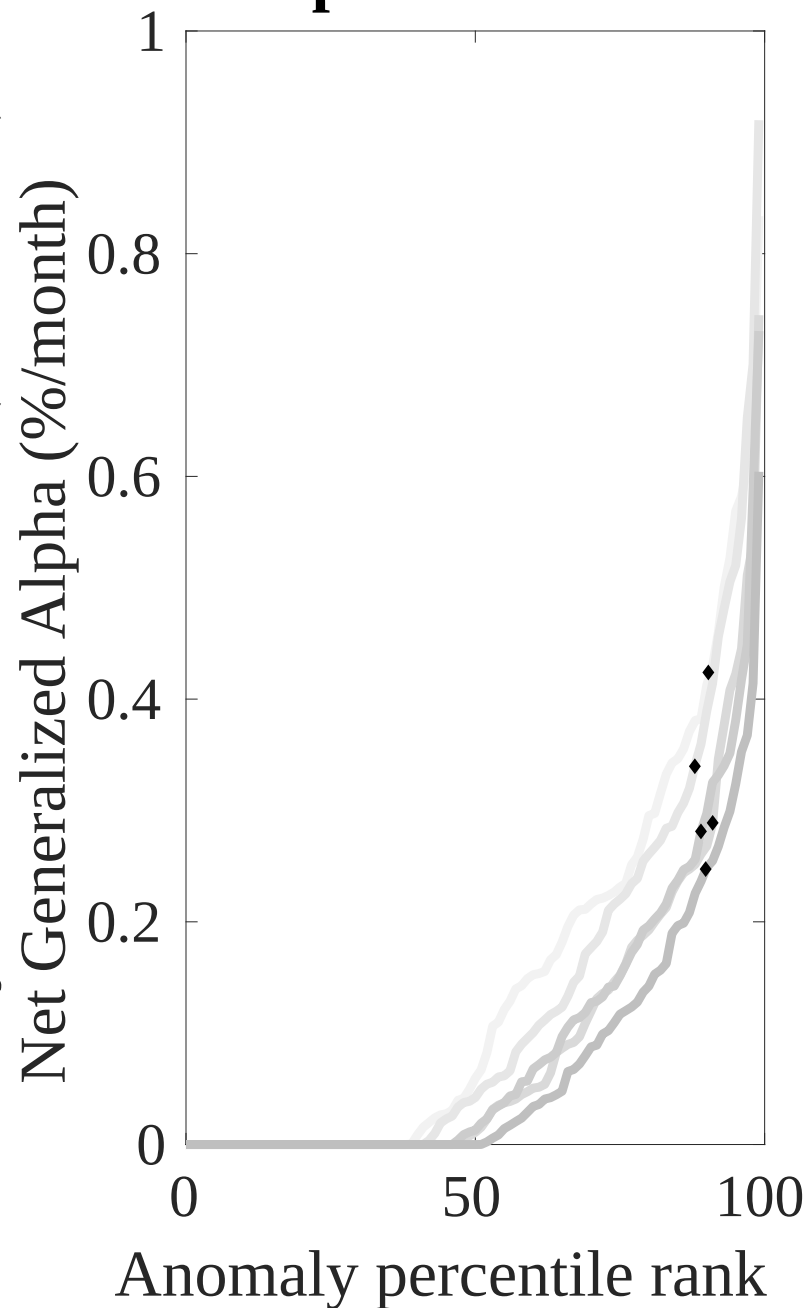
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the IOE trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



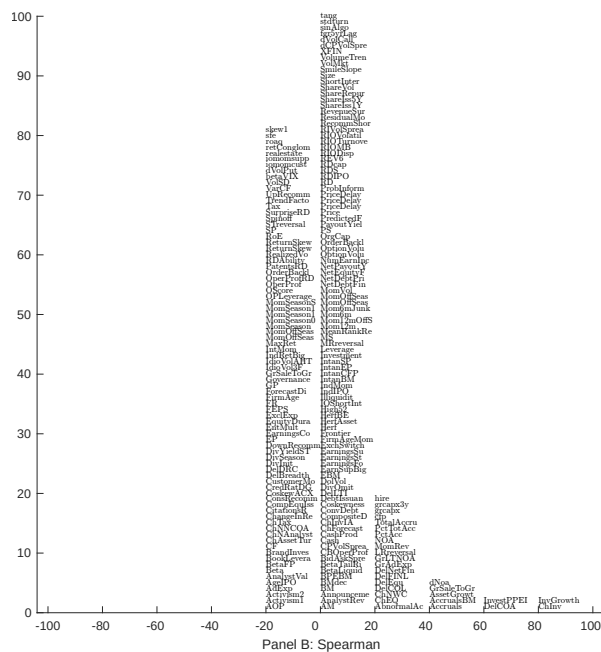
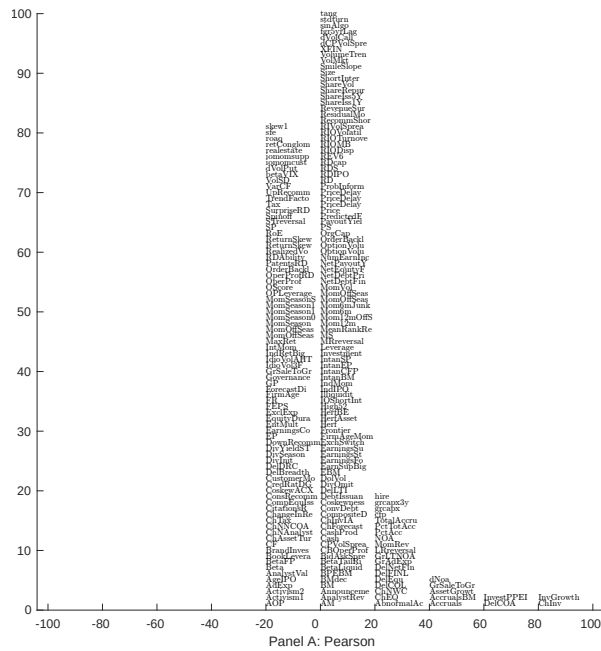


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 209 filtered anomaly signals with IOE. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

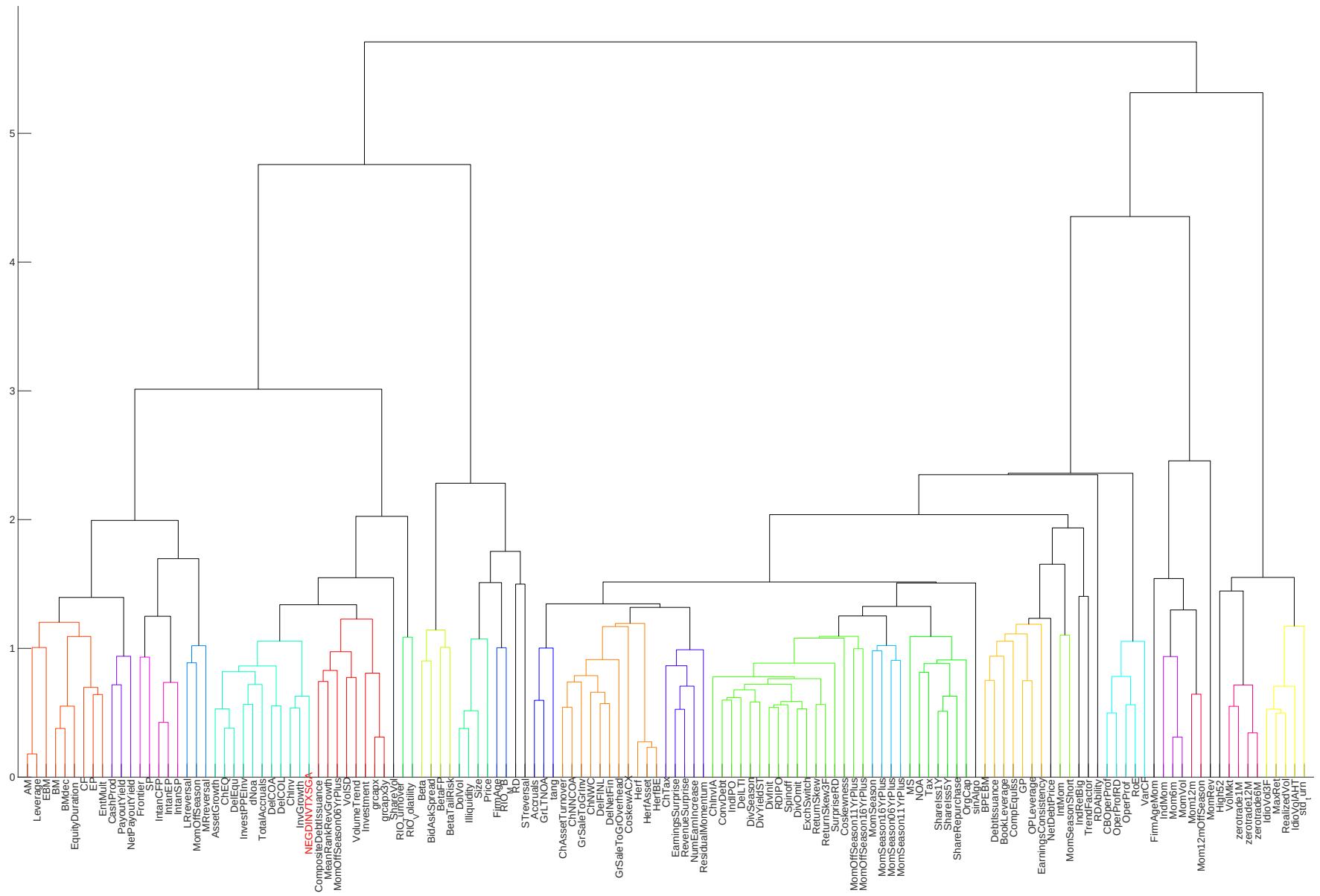


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

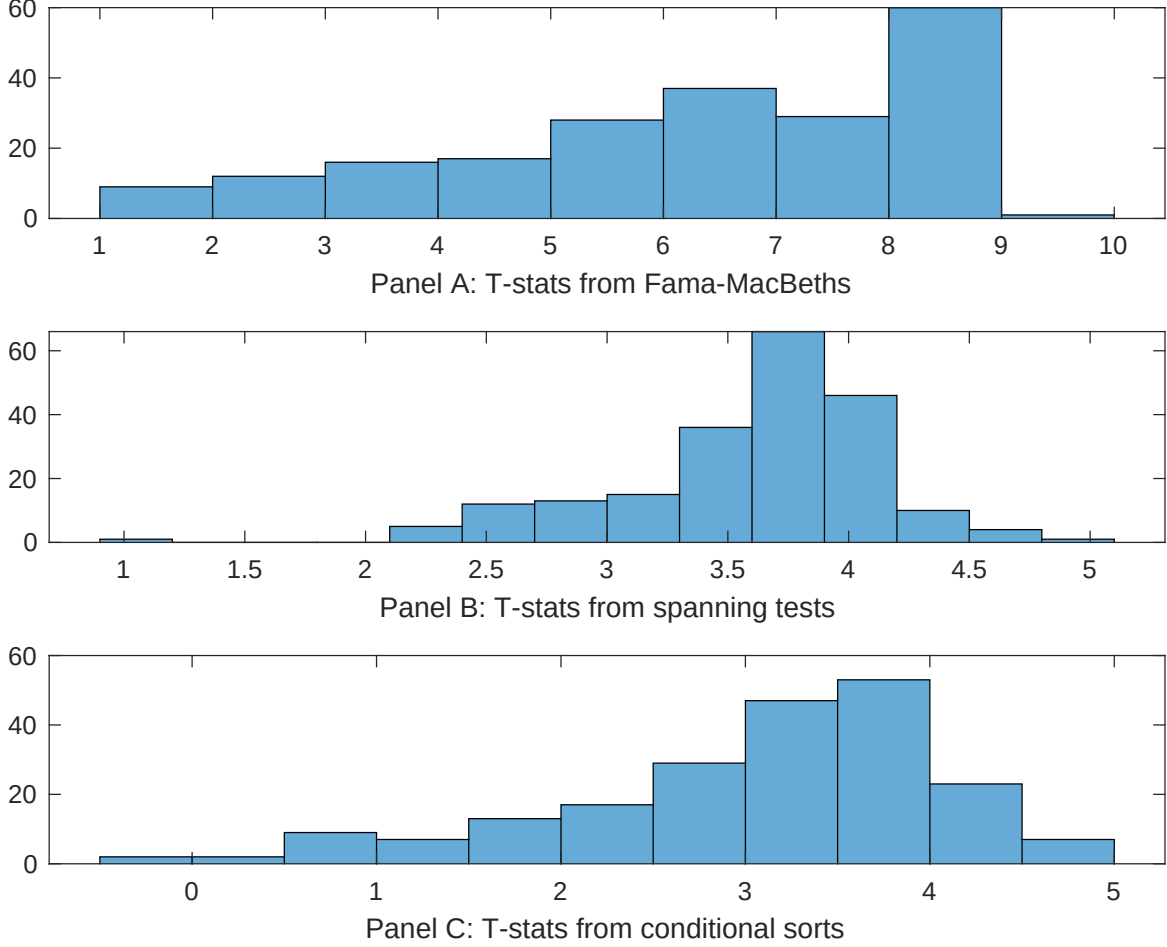


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of IOE conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IOE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IOE}IOE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IOE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on IOE. Stocks are finally grouped into five IOE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IOE trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on IOE. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{IOE}IOE_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Inventory Growth, Inventory Growth, change in ppe and inv/assets, Change in current operating assets, Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [5.73]	0.13 [5.67]	0.14 [5.96]	0.13 [5.79]	0.14 [6.16]	0.13 [5.55]	0.14 [5.88]
IOE	0.22 [4.02]	0.21 [3.45]	0.95 [1.82]	0.23 [5.08]	0.17 [3.85]	0.28 [6.16]	0.83 [1.21]
Anomaly 1	0.25 [4.29]						0.18 [0.20]
Anomaly 2		0.43 [6.88]					0.48 [0.06]
Anomaly 3			0.16 [7.65]				0.60 [1.93]
Anomaly 4				0.18 [5.92]			0.33 [0.92]
Anomaly 5					0.96 [8.14]		0.63 [4.38]
Anomaly 6						0.80 [5.55]	-0.10 [-0.77]
# months	732	732	732	732	732	713	713
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the IOE trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{IOE} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Inventory Growth, Inventory Growth, change in ppe and inv/assets, Change in current operating assets, Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14 [1.85]	0.25 [3.26]	0.24 [2.86]	0.25 [2.92]	0.24 [2.77]	0.31 [3.50]	0.18 [2.38]
Anomaly 1	53.61 [14.10]						33.78 [7.15]
Anomaly 2		44.95 [14.60]					30.90 [8.15]
Anomaly 3			21.74 [5.46]				3.63 [0.97]
Anomaly 4				23.66 [5.69]			1.96 [0.48]
Anomaly 5					11.69 [2.27]		-0.16 [-0.03]
Anomaly 6						15.30 [3.24]	-11.85 [-2.65]
mkt	-0.28 [-0.15]	-2.62 [-1.44]	-2.24 [-1.10]	-2.13 [-1.05]	-1.96 [-0.95]	-1.71 [-0.83]	-1.70 [-0.98]
smb	3.81 [1.42]	0.33 [0.12]	-3.51 [-1.19]	2.29 [0.75]	-3.58 [-1.19]	-2.20 [-0.74]	2.82 [1.05]
hml	-6.07 [-1.71]	-4.76 [-1.35]	-4.86 [-1.23]	-11.38 [-2.69]	-2.86 [-0.72]	-4.67 [-1.15]	-4.63 [-1.26]
rmw	-7.70 [-2.11]	-11.54 [-3.23]	-16.99 [-4.28]	-14.10 [-3.53]	-17.03 [-4.22]	-16.34 [-4.08]	-7.30 [-2.12]
cma	35.38 [6.12]	33.59 [5.83]	53.80 [8.24]	57.51 [9.28]	56.38 [6.59]	55.52 [7.74]	27.71 [3.60]
umd	5.51 [3.00]	3.91 [2.12]	8.73 [4.33]	9.76 [4.83]	9.13 [4.44]	8.11 [3.95]	4.28 [2.40]
# months	732	732	732	732	732	713	713
$\bar{R}^2(\%)$	47	47	35	35	32	33	52

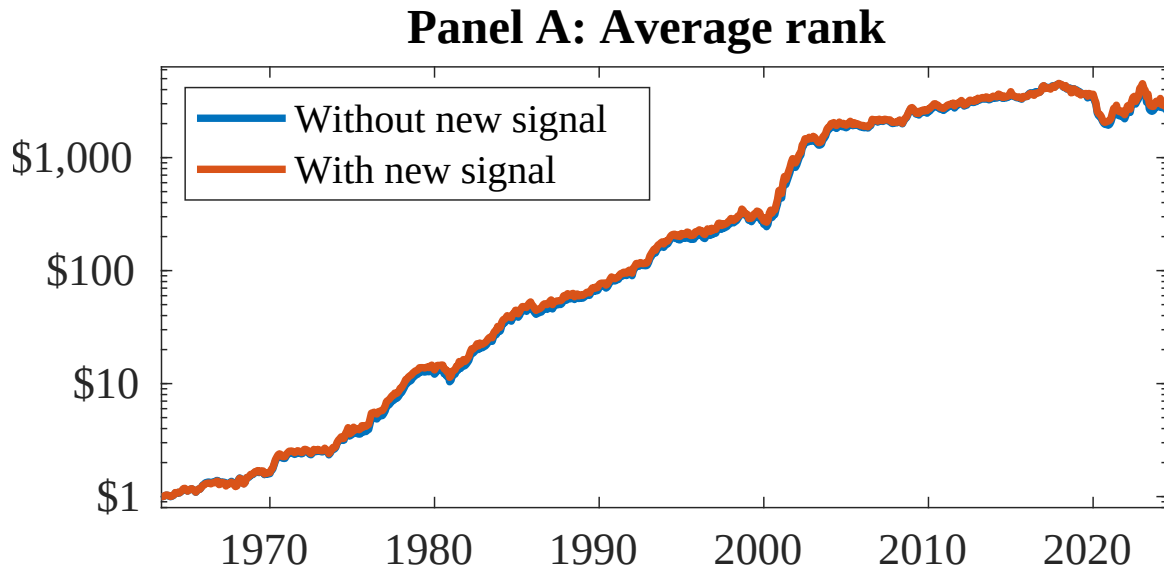


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as IOE. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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